



MINING DEALER BEST PRACTICE

Hydraulic Cylinder Head Salvage by Installing Bronze Bushing

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

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1.0 Introduction

During the operation of hydraulic cylinders, burr scratches may appear in heads of hydraulic cylinders. Previously, those heads were replaced with the new ones. Now Borusan Makina Kazakhstan salvages hydraulic cylinder heads by installing bronze bushings. It enables to significantly reduce the repair cost of hydraulic cylinders.

This method of hydraulic cylinder head salvage is applicable for any big hydraulic cylinders on condition that the place where the bronze bushing is installed should not have any grooves for seals, because the bronze bushing serves as the guide there.

2.0 Best Practice Description

In order to salvage hydraulic cylinder heads, bronze bushings are purchased from local suppliers. Then those bushings are installed to the heads and bored to the necessary sizes according to drawings. Thus, hydraulic cylinder heads are salvaged but not replaced that allows saving money on overhauls of hydraulic cylinders because the salvaged cylinder head provides a 31.5% cost saving compared to the new one.

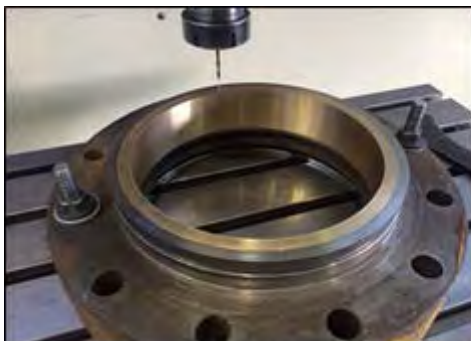
3.0 Implementation Steps

As an example, 777F Truck Body Lift/ Hoist Cylinder Head salvage by installing the bronze bushing is shown below.

- 1) The damaged area in the hydraulic cylinder head is bored.



- 2) The bronze bushing is installed into the cylinder head. In the edge of the joint it needs to drill 3 x holes with the depth of 12-15 mm and at the distance of 120° from each other.



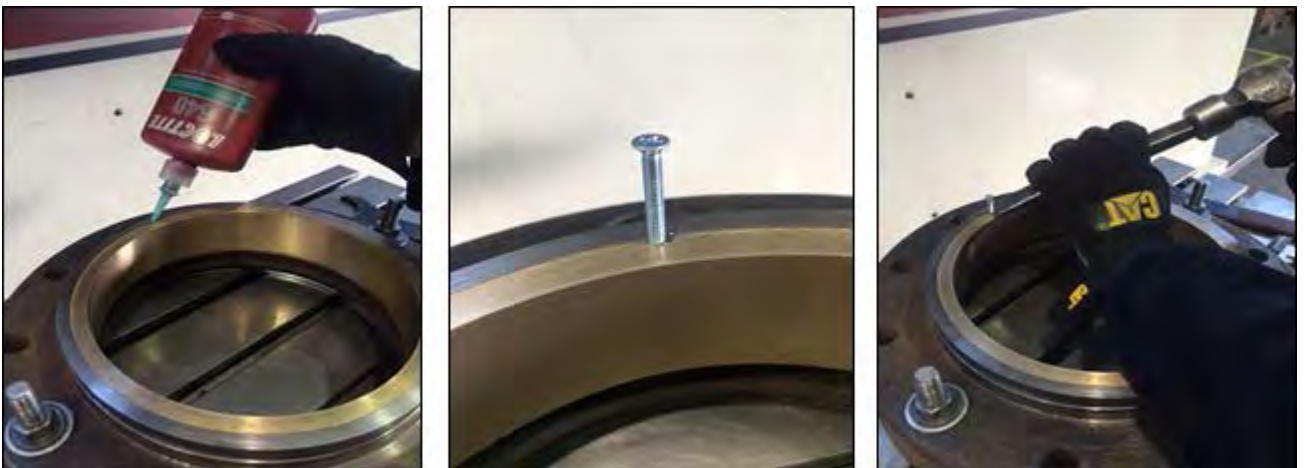
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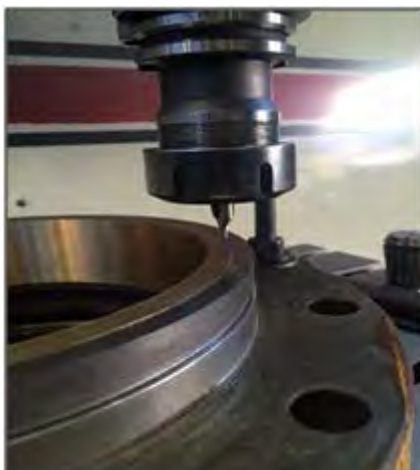
- 3) The thread is tapped in those holes.



- 4) 3 x toggle bolts with Loctite 640 are installed in the holes. The remaining parts of the bolts a



- 5) Then the internal part of the bronze bushing is bored to the necessary size as per the technical drawing.



The cylinder head has been salvaged by installing the bronze bushing

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4.0 Benefits

- Hydraulic cylinder overhaul cost has been reduced (the salvaged cylinder head gives a 31.5% cost saving compared to the new one).

5.0 Resources Required

- Machine Shop's equipment (milling machine, etc.);
- Bronze bushings purchased from local suppliers;
- Toggle Bolts;
- Loctite 640.

6.0 Supporting Attachments / References

- Bushing information – The diameter is 224.15 -0.05mm. The tension is about 0.1mm. Before pressing, the bushing is cooled down to -50 degrees.
- Highest hours of cylinders operating in the field with this rework. – It works up to the next overhaul (12,000 hrs.)
- No failures to date. We started to use this procedure a long time ago (more than 1 year). Now this procedure has become the standard job.
- At the time of the next rebuild, do you need to replace the bushing due to wear? – If the burrs are revealed in the bushing during inspection, it is replaced. If not, it's not replaced.

7.0 Related Best Practices

N/A.

8.0 Acknowledgements

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